

LungVision-AI: Object Detection for Lung Cancer Localization in CT Scan Images

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Abstract

Lung cancer remains one of the leading causes of cancer deaths worldwide, for which improved diagnostic imaging technology for early detection is needed. This paper presents LungVision-AI, an artificial intelligence system based on YOLOv5 object detection models, to assist radiologists in localizing potential lung cancer nodules from CT scan images. Three variants of YOLOv5—YOLOv5mu, YOLOv5lu, and YOLOv5m6u—were trained and tested on a 13,237 annotated dataset of CT scans, augmented using preprocessing and augmentation methods to make models robust. Precision, recall, and mAP values were utilized to compare models. Usability testing with a medical radiologist revealed an 80% detection success rate, with LungVision-AI performance in nodule detection, albeit with still challenges in dense anatomical areas. Real-world deployment was made through a web application, with real-time predictions using an intuitive interface. Outcomes indicate the promise of LungVision-AI to improve diagnostic efficiency and accuracy, with scope for future clinical workflow integration after optimization and explainability improvement.

CCS Concepts • AI in Healthcare • Computer Vision • Machine Learning • Artificial Intelligence

Additional Keywords and Phrases: YOLOv5, Lung Cancer Detection, Medical Imaging, CT scans, Medical AI Tools, Radiology, Real-time Localization.

I. Introduction

Lung cancer represents the most prominent global challenge in health, next after breast cancer as the second most diagnosed cancer, and is considered the leading cause of all cancer-related deaths, with more than 25% of deaths due to lung cancer. Between 2.2 million new cases and 1.8 million deaths were attributed in 2020 to lung cancers [\[1\]](#). Among its types, **non-small-cell lung cancer** (NSCLC) is probably 85-90%. Even with the significant advancement in treatment strategies such as nanomedicine and photothermal therapy, survival is still below expectations, thus creating an urgent need for more significant interventions in terms of diagnosis and treatment approaches [\[2\]](#), [\[3\]](#).

Recent advancements in artificial intelligence, particularly with deep learning algorithms as demonstrated by **Convolutional Neural Networks** (CNNs), can be said to offer promise in augmenting lung cancer detection. Such AI systems, with the support of **Computer-Aided Detection** (CADe), improve sensitivity while reducing false positives, and innovations such as radiomics classify lung nodules as malignant or benign. These AI applications are particularly beneficial in combating the problems arising from the enormity of lung cancer heterogeneity. However, reproducibility issues and generalizability of AI models over multiple imaging protocols persist [\[5\]](#), [\[6\]](#), [\[8\]](#), [\[11\]](#), [\[12\]](#).

This study's objective is to develop an AI-based tool that applies **YOLOv5** to assist radiologists in identifying any suspect regions of lung cancer in CT scan images. This tool strives to optimize diagnostic speed and accuracy as it represents a real gap in diagnostic tools that either achieve speed without accuracy or accuracy and compromise speed.

Existing literature highlights many gaps in AI-based lung cancer detection, some of which are limited access to multimodal imaging, difficulty in identifying small nodules, slow processing speed, dense lung areas, and the absence of model interpretability. Multimodal imaging is not available at all health developments; hence, it cannot be utilized in a lot of clinical settings [\[11\]](#), [\[17\]](#). Small nodules appear like their surrounding tissue, and therefore detection rates become lower [\[11\]](#), [\[13\]](#), [\[16\]](#), [\[17\]](#). With slow processing speeds, real-time application becomes difficult [\[14\]](#)-[\[16\]](#), while dense lung areas complicate accurate detection [\[13\]](#). Moreover, interpretability of AI models is ordinarily a gap in clinical practice; hence, results do not get construed into clinical trust [\[11\]](#), [\[13\]](#), [\[14\]](#), [\[17\]](#).

This paper compares three variants of YOLOv5, which are YOLOv5mu (medium ultra), YOLOv5lu (large ultra), and YOLOv5m6u (medium 6 ultra). These variants were utilized since they offer good trade-offs between accuracy and speed for medical image tasks. YOLOv5mu is a lighter and less resource-hungry model that lowers the

computation costs and is thus best utilized in real-time or computation-limited applications. YOLOv5lu provides increased accuracy but at the cost of slower computation and higher computation needs. YOLOv5m6u, on the other hand, balances the YOLOv5m series improvements and the "6" improvements (longer layers and larger input), providing better accuracy (mAP50-95: 53.6%) but at the cost of higher computation needs (FLOPs: 65.7 billion). All three variants were trained on 13,237 lung CT scans annotated, preprocessed, and augmented for stability. Performance metrics like Mean Average Precision at 50% IoU (mAP50), Mean Average Precision across IoU thresholds (mAP50-95), Precision, Recall, and a set of Loss Metrics were used to compare the performance of the models. The findings indicate that YOLOv5 architectures with multi-scale feature fusion are very efficient and accurate in processing [\[19\]](#), and as such, they are very commendable for early detection of lung cancer development [\[17\]](#).

The significance of this research lies in its potential to revolutionize lung cancer detection. This study would very much decrease the delay and improve the patient's condition by offering an AI-powered tool that brings time efficiency with accuracy into diagnosis. Integrating such a tool into the clinical workflow would lessen the burden on radiologists and allow them to devote their time to critical decision-making tasks. Future developments from this research will pave the way for more personalized and effective approaches to lung cancer management, ultimately resulting in increases in survival rates and quality of care.

II. Theoretical Background

Artificial intelligence has made great strides in medical imaging concerning the early detection of lung cancer, especially concerning critical lung nodules that are often challenging to detect manually [\[10\]](#). Deep learning techniques, such as Convolutional Neural Networks (CNN), have been adopted directly into healthcare workflows to enhance the sensitivity of Computer-Aided Detection (CADe) systems that assist radiologists in reducing false positives and improving detection efficiency, especially in a population at elevated risk for lung cancer screening [\[5\]](#), [\[8\]](#). Radiomics is the most recent development in AI and combines all these techniques that explore the quantitative features from medical images, such as shape, texture, intensity, etc., to categorize lung nodules into either malignant or benign [\[6\]](#), [\[12\]](#). Despite its promise, reproducibility of radiomics using an entirely different imaging protocol still needs to be improved and further integrated with deep learning models to improve its robustness and accuracy [\[6\]](#), [\[12\]](#).

Low-dose computed tomography (CT) is the gold standard for early detection of lung cancer, and this detection process has shown great promise in being significantly improved through the application of AI systems [\[9\]](#). For instance, models used in the LUNA16 challenge have exceeded manual evaluations in detecting lung nodules, besides reducing false positives [\[9\]](#), [\[12\]](#). These tools could improve the survival rate of patients when intervention based on these tools occurs earlier than standard care methods [\[10\]](#). Deep learning models can accurately locate, segment, and classify pulmonary nodules in even the earliest stages, when small, difficult to detect manually [\[12\]](#).

This is further supported by AI-based segmentation of lung nodules. Another important application of AI is the segmentation of lung nodules. High-profile deep learning models such as U-Net and SegU-Net have shown superior results to conventional approaches for the accurate delineation of nodules from the surrounding tissues, enabling their automatic volume calculation [9]. Accurate segmentations are vital for early diagnosis and determination of treatment strategies, since nodule volume estimations will aid in the determination of appropriate action [9], [12].

Building on these advancements in detection and segmentation, AI systems are now beginning to approximate the level of using scientific knowledge in predicting outcome and treatment response for patients [14]. Virtual biopsies, imaging-based methods that are invasive, also include tools for diagnosing and predicting the type of lung cancer and responses to therapy [7]. Biomarkers, such as epidermal growth factor receptor (EGFR) mutations, can be detected using AI, which influences the treatment plan [5]. Moreover, through the AI model, patients can be stratified according to the prediction of their responses to immunotherapy and other treatments, which leads to more personalized care [5]. Integrating the AI revolution into screening and diagnosis of lung cancer further lightens the burden of radiologists and increases their diagnostic accuracy, especially in large screening programs [5], [7]. They can convert huge amounts of screening data into consistent and objective analyses [10]. AI tools are being used in diagnostic and therapeutic decision-making including ongoing developments in predictive models on treatment responses [5], [6].

Extending the application of AI in lung cancer detection, real-time models like YOLO offer transformative possibilities for medical imaging. Great real-time performance of YOLO also extends to multiple quick and accurate object detection which makes it amenable for medical applications where fast and accurate identification of the variety of objects such as nodules or anomalies is of importance. This single-stage architecture enables the effective detection that is always a necessity in time-bound medical applications [15]. The speed and efficiency of YOLO make it apt for the detection of lung cancer nodules especially in the initial stages when these nodules are small enough to be difficult to detect. Advancement like Lung-YOLO empowers real-time processing capabilities of YOLO enhanced feature extraction, resulting in fast and accurate detection of small irregular lung nodules which is important for early diagnosis leading to better patient outcome [16].

Given these capabilities, the selection of YOLOv5 for lung cancer localization aligns seamlessly with the need for real-time detection and efficient processing of multimodal medical images, including PET, CT, and PET/CT. Based on the performance of YOLO and recommendations from our research adviser, YOLOv5 was selected as the model for lung cancer localization, as it aligns well with the requirements for real-time detection and efficient processing of multimodal medical images, including PET, CT, and PET/CT [15]-[17]. The YOLOv5 version suits perfectly for lung cancer imaging since it is real-time in the detection process and is efficient in processing multimodal medical images, including PET, CT, and PET/CT. Another advantage of quick identification and localization of small lesions from complex data coupled with features such as the Global-Local Feature Interaction Module and Cross-Scale Attention Fusion adds to the detection of lung tumors more accurately and quickly. Thus, YOLOv5 becomes especially effective for early diagnosis, where precision matters. [17]

In order to further optimize YOLOv5 usage in the context of lung cancer detection, this study evaluates three variants of YOLOv5, namely; YOLOv5mu (medium ultra), YOLOv5lu (large ultra), and YOLOv5m6u (medium 6 ultra). Selection of model variants is based on balancing speed, computational efficiency, and accuracy in detecting lung cancer. Provided with faster inference time and lesser computational needs (233.9 milliseconds and 64.2 billion FLOPs), YOLOv5mu makes it highly suitable for real-time or resource-constrained environments. The model YOLOv5lu offers a better mAP_{val50-95} (Mean Average Precision) accuracy of 52.2% but incurs a slower processing time of 408.4 milliseconds and requires a significantly greater amount of resource at 135.0 billion FLOPs [18]. Meanwhile, YOLOv5m6u provides a balance between both, by inputting a larger size of input dimensions (1280×1280), and attaining a higher mAP_{val50-95} score (53.6%) with an increase in processing time (810.9 milliseconds) and 65.7 billion FLOPs. All three models were independently trained on 13,237 annotated lung CT images and incorporated multi-scale feature fusion to enhance robust detection in particular for small lung nodules which are critical for early diagnosis [19]. This will allow comparative evaluation among the models on where each shines or falters, comparative insights into the models' suitability to various medical imaging applications according to the specific requirements speed, available resources, and diagnostic accuracy.

As it evolves, AI will acquire a greater role in management for lung cancer patients, by offering a more personalized approach to treatment [9]. It has already enhanced detection and classification of nodules, identification of key biomarkers, and optimization of treatment plans. Through improved operational efficiency, AI can increase lung cancer management effectiveness and thus improve future patient outcomes [11].

III. Review of Related Literature

This section examines current research on lung cancer detection through AI-based tools, identifying critical key gaps in these studies and exploring how LungVision-AI could serve as a potential solution to bridge these gaps. The review discusses challenges in related literature like detecting small nodules, challenges in detecting anomalies in dense or complex lung areas, and lack of model interpretability which hinders clinical trust.

A. Difficulty Detecting Small Nodules

Two studies about *Artificial intelligence and lung Cancer: Impact on improving patient outcomes* and *Artificial intelligence: A critical review of applications for lung nodule and lung cancer* have focused on putting AI to work for furthering the lung cancer detection through machine-learning and deep-learning algorithms such as convolutional neural networks (CNN), which are specifically focused on detection, classification of lung nodules, determining benign or malignant nature, and biomarker integration that provides exhaustive diagnostic evaluation [11], [13]. Key findings demonstrate effective AI algorithms to compute traces of human-like or even a better

performance when compared to radiologists with such model innovations as *Lung-YOLO* and *CCGL-YOLOV5*, where achieved high mAP (Mean Average Precision) scores of 97.9% and better differentiation in multimodal feature extractions in small nodule detection processes [16], [17]. However, some shortcomings still exist from the aforementioned studies, including reduced output across datasets of very limited applicability, being "black box" tools that are not interpretable, and limited detection ability of small nodules with low contrast and high false-positive rates, especially in difficult imaging conditions [13], [16], [17].

Building on these advancements, according to studies titled: *Lung nodule detection in medical images based on improved YOLOV5S* and *An improved YOLOv5 network for lung nodule detection*, YOLOv5 is a very competent model for lung cancer detection with its remarkable speed, accuracy, and scalability, making it the right solution for real-time clinical applications [20], [21]. In fact, several medical imaging modalities, such as CT and X-ray scans, form smaller pieces called grid cells wherein careful predictions of class probabilities and bounding boxes for detection can be made processing through convolutional neural networks (CNN) using YOLOv5. It enables efficient and precise detection of small lung nodules which are important for early diagnosis, as seen in [20]. Improvements like YOLOv5mu and YOLOv5lu target the actual problem of nodule detection below 9 mm, showing an edge over detection methods in adverse contrast and challenging imaging conditions by using advanced object detection systems like flexible anchor-free detection. All these advancements reassure that higher accuracy, sensitivity, and lower false-positive rates would be secured, thus creating a perfect bridge between the old and the innovative approaches by using YOLOv5 and its sub-groups as powerful platforms in the fight against old methods of small lung nodule detection [18].

B. Challenges in Dense or Complex Lung Areas

Previous studies on AI-assisted lung cancer detection have focused on enhancing nodule detection through imaging systems such as low-dose computed tomography (CT) scans, chest X-rays, and the integration of clinical data. The objectives include improving diagnostic sensitivity and specificity, minimizing false positives, and integrating clinical features such as tumor mutations to aid in therapeutic decision-making. For instance, models such as those in *Artificial intelligence and lung Cancer: Impact on improving patient outcomes* have aimed at developing deep learning techniques to improve lung cancer diagnosis through imaging analysis. Other goals included better identification of malignant nodules in complex lung regions [11]. The works of the study: *Artificial intelligence: A critical review of applications for lung nodule and lung cancer* further supported the use of convolutional neural networks (CNNs) in improving lung nodule detection and lung cancer risk prediction, with results such as AUC values of 0.94 and 0.955, respectively, demonstrating good sensitivity and specificity [13]. Additionally, the study about the *Application of artificial intelligence in lung cancer* stated that AI systems have been shown to outperform radiologists in detecting lung nodules, achieving sensitivities as high as 90%, and predicting molecular characteristics of tumors for personalized treatment planning [14]. Despite these advancements, limitations remain, particularly in detecting

lung cancer in denser and anatomically complex areas, where nodules are often obscured by overlapping structures [11], [13], [14].

To address these limitations, according to studies titled: Lung nodule detection in medical images based on improved YOLOV5S and An improved YOLOv5 network for lung nodule detection, models such as YOLOv5 exhibit very promising performance concerning lung cancer detection, especially in dense areas of the lungs. The lightweight nature transports YOLOv5 real-time detectors, thus becoming a perfect device for small nodule detection with multi-scale detection techniques, giving a speed of around 140 FPS with a low model size. Although applications in the pure sense of YOLOv5 become a challenge in some complex cases, notable advancements in detecting small or hidden nodules are achieved with improvements such as attention mechanisms and feature fusion [20], [21]. The newly emerging versions, YOLOv5mu, YOLOv5lu, and YOLOv5m6u, increase the detection ability for pulmonary nodules particularly in denser and anatomically complex lung zones with respective mAP (Mean Average Precision) scores of 49.0, 52.2, and 53.6. The optimizations allow a balance in speed and accuracy: YOLOv5mu runs in 233.9 milliseconds, YOLOv5lu in 408.4 milliseconds, and YOLOv5m6u in 810.9 milliseconds on CPU (ONNX). The large parameter settings and high FLOPs contribute to complex feature extraction necessary for accurate lung cancer diagnosis, while their anchor-free detection ability provides flexibility in detecting nodules of various sizes and locations [18].

C. Trust and Interpretability Obstacles

The issue of human trust in AI systems has been a focal point of research, particularly in the context of interpretability and feedback. Both studies focus on the factors that affect AI trust, but the findings are different. A study about *“Will we trust what we don’t understand? Impact of model interpretability and outcome feedback on trust in AI”* examines the effect of interpretability and outcome feedback on trust and human performance in AI-assisted tasks. The experiment showed that interpretability had very little impact on trust, whereas outcome feedback had a strong effect on trust, although both factors had modest effects on task performance [26]. This suggests that while interpretability is often emphasized in AI development, feedback mechanisms may play a more significant role in building trust. The study about *“Trust in AI: Interpretability is not necessary or sufficient, while black-box interaction is necessary and sufficient”* further details this by suggesting that interpretability is neither necessary nor sufficient for trust. Instead, it advocates for a “black-box interaction” approach, where the ability to run AI models at-will is key to building trust [27]. The previously mentioned study also critiques the centrality of interpretability, suggesting that understanding an AI’s behavior through behavior certificates, rather than its internal workings, is more effective for fostering trust. Both studies point to the value of human interaction with AI models but differ in placing more emphasis on interpretability over practical interaction for building trust. The limitation with both studies is that modest improvements in task performance resulting from interpretability indicate that trust does not always seem to relate to better results in tasks assisted by AI. Similarly, in medical imaging, AI and deep learning techniques are being used

to improve lung cancer detection and classification. Though such models have been very successful, with high AUC values and even surpassing the performance of radiologists in certain phases, challenges remain, especially with model interpretability. As mentioned in the study of *Artificial intelligence: A critical review of applications for lung nodule and lung cancer*, this issue impacts clinical trust and integration of AI models into practice since the black-box nature of many deep learning algorithms makes their generalization complicated, and there are concerns regarding external validation in diverse populations [\[13\]](#).

The primary gaps highlighted from the previous literature about trust and interpretability in AI models for lung cancer detection, especially black-box nature of deep learning models, are addressed through advancements in YOLOv5. The default YOLOv5 model is the one with a balance of performance and efficiency, suitable for the detection of lung nodules in medical images. Since YOLOv5 does not inherently provide interpretability, real-time detection capabilities and fast processing allow healthcare professionals to depend on it for rapid decision-making. This object detection framework of the model can detect small and diverse lung nodules for early detection. Although its decision-making process remains difficult to interpret, consistent performance and accuracy in detecting lung nodules in complex datasets help build trust in its utility for clinical applications. As emphasized by the studies on *Lung nodule detection in medical images based on improved YOLOV5S* and *An improved YOLOv5 network for lung nodule detection*, this reliability, combined with further research on interpretability, could enhance AI integration into medical practice and foster greater trust among healthcare professionals [\[20\]](#), [\[21\]](#).

D. Understanding Different Findings in the Literature

The review of related literature highlights both alignment and divergence with existing studies, informing the direction of LungVision-AI. Prior research agrees on the efficacy of YOLOv5 in detecting small nodules and navigating dense lung regions, emphasizing its speed, scalability, and accuracy for early diagnosis. However, discrepancies arise regarding the necessity of model interpretability for clinical trust, with some studies prioritizing consistent performance over transparency. Our study addresses these gaps by enhancing YOLOv5 with attention mechanisms and anchor-free detection to improve accuracy in low-contrast conditions while leveraging its real-time capabilities to build trust through reliability. Additionally, we propose integrating explainable AI (XAI) features to offer clinicians contextual insights into decision-making processes, bridging the gap between performance and interpretability to foster greater adoption in medical practice.

IV. Methodology

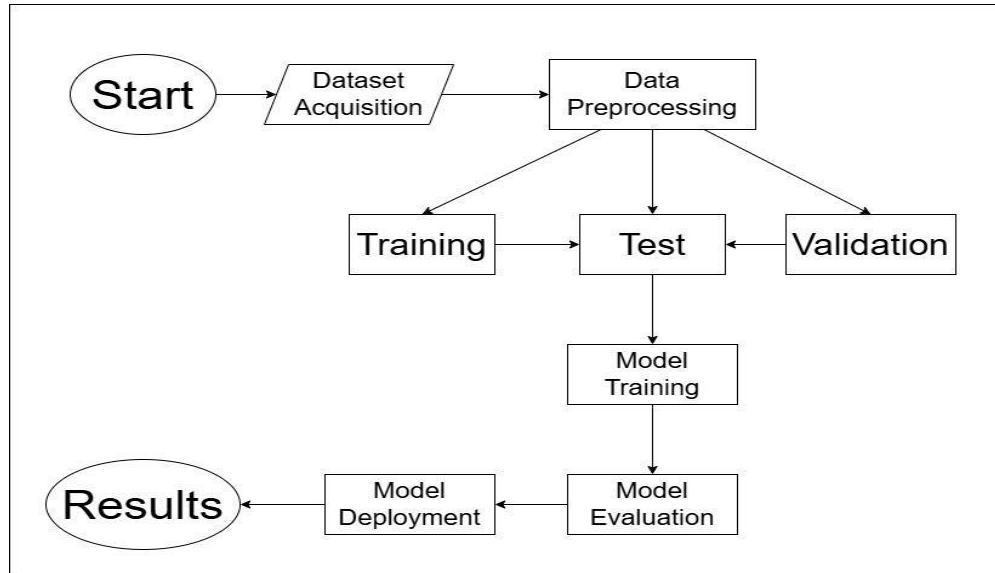


Figure 1: Methodology Flowchart

A. Data Acquisition

Data for this study was sourced from The Cancer Imaging Archive (TCIA). This service contains an extensive archive of public and private medical cancer images, which is available for download and includes a collection of computed tomography (CT) scans from 355 patients. The images were obtained from patients suspected of having lung cancer. Five academic thoracic radiologists specializing in lung cancer annotated the locations of each tumor. Two radiologists had more than 15 years of experience, while the rest had more than 5 years. After one of the radiologists labeled each subject, the other four radiologists conducted a verification, resulting in all five radiologists reviewing each annotation file in the dataset. The dataset will be divided into three subsets for the model's training: 70% will be for training, 15% for testing, and 15% for validation[4].

B. Data Preprocessing and Augmentation

Several preprocessing steps were used to guarantee consistency and enhance the model's performance[5],[13],[14]. Orientation was applied to all images to correct alignment distinctions, ensuring uniformity across the dataset. The next step is resizing images into 640x640 pixels to ensure compatibility with the YOLOv5

architecture[14]. All images were converted to grayscale to highlight the features of interest, reduce color variability, and reduce data input for the model.

Four different augmentations have been performed on each image, which is pivotal in enhancing model robustness. The first variety involved fixed rotation, counterclockwise and upside down, to make the model recognize from various orientations and improve its ability to recognize from different angles. The second augmentation includes brightness changes from -15% to +15%, which simulate different light conditions. The third phase is blurring: a blurring kernel of up to 1.5 pixels is used on the images to imitate emphasis variation. Random noise was added to as much as 1.96% of the pixels within each photograph to simulate real-world artifacts such as speckles or sensor noises[14]. All preprocessing and augmentation steps jointly improve the dataset, enabling the model to learn well and generalize better for unseen data.

C. Model Architecture

The models for object detection in this study are centered on the architecture of YOLOv5[18],[21]. The three main components constituting YOLOv5 are the Backbone, Neck, and Head[22]:

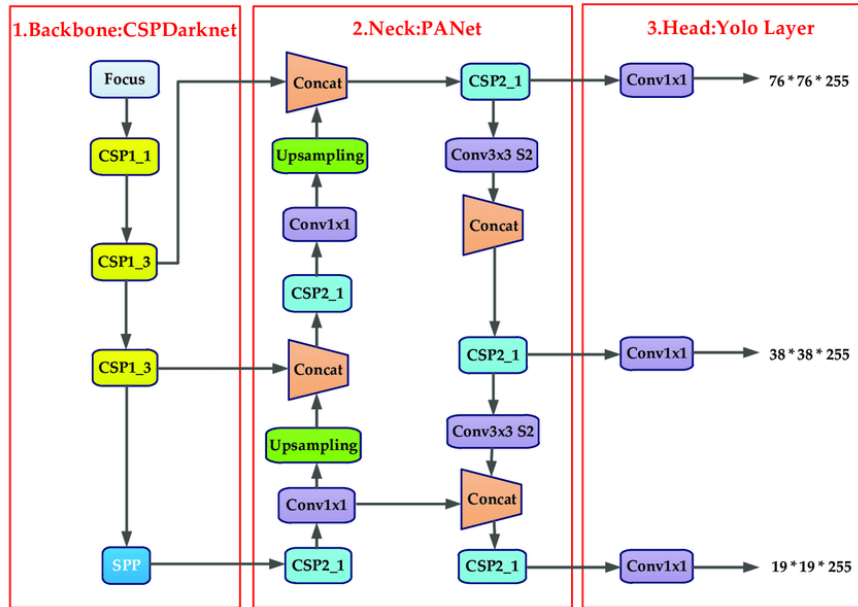


Figure 2: The network architecture of YOLOv5[25]

Backbone: Utilizes the use of CSP (Cross-Stage Partial) to reduce computational expenses and model complexity by splitting feature maps into two parts and merging back while not compromising its performance. The efficiency improvements are relevant for the YOLO series, as speedier inference and small model size are required[22].

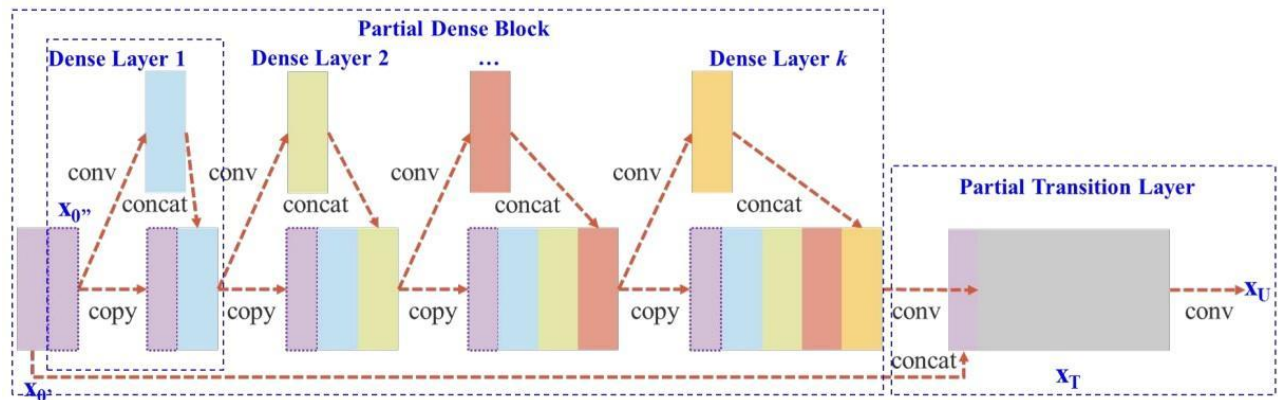


Figure 3: CSPDenseNet[24]

Neck: YOLOv5 uses a Path Aggregation Network (PA-Net) for feature aggregation to enhance the assimilation of information from multiple layers sourced from the CSP backbone. The design was based on EfficientDet's BiFPN, a model that maximizes feature fusion[22].

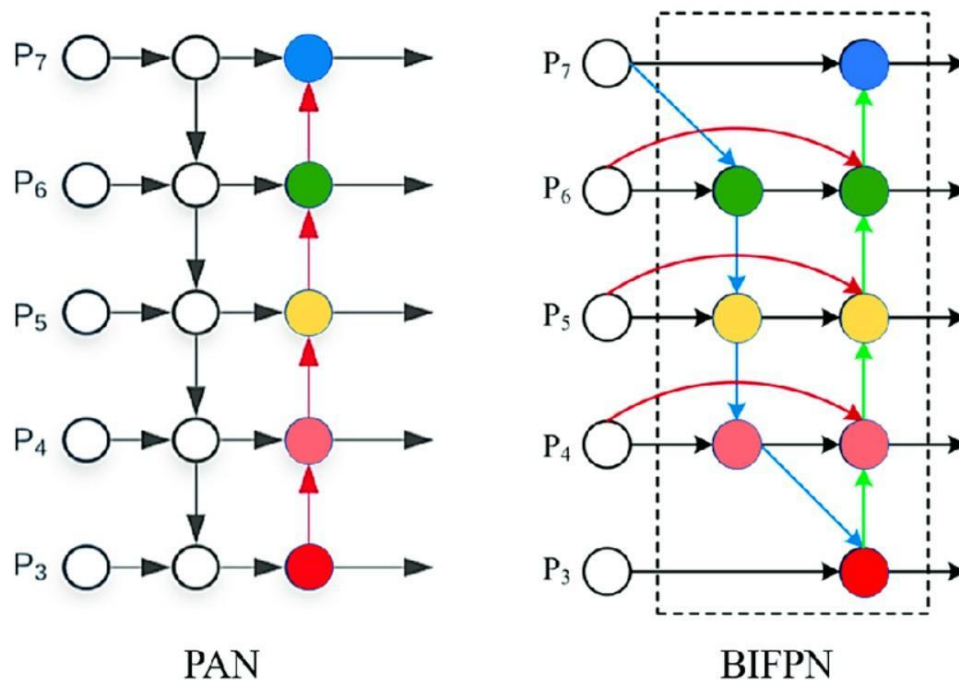


Figure 4: The comparison of PANet and BiFPN[23]

Head: The head is where the final predictions will be made. It has several detection layers that make predictions on bounding boxes, class probabilities, and objectness scores for each detected object. YOLOv5 generates outputs on three different scales for prediction, enabling the model to detect small, medium, and large objects effectively.[\[25\]](#).

The whole architecture of YOLOv5 is designed to deliver high performance in speed and accuracy, which makes it useful for real-time applications in many fields, including medical imaging.[\[21\]](#)

D. Model Training

In this research, an object detection model was developed using YOLOv5. The first step is to collect diverse data and annotate it with the correct bounding boxes. Accurate labels are necessary in order to train well. The data is augmented with different lighting, angles, and even dimensions to improve the robustness of the model. We trained YOLOv5 using the annotated dataset, tuning the parameters to minimize the loss function and calculating how well the bounding box predictions come close to the actual object boxes. Various evaluative metrics, such as mean Average Precision (mAP) and Intersection over Union (IOU), were applied during training to determine the model's performance. Thus, it could be polished for the performance optimization on the accuracy and generalization of the model in a way that suits the application in real-time with the model.[\[17\]](#),[\[19\]](#)-[\[21\]](#). Upon completion of the data collection, preprocessing followed, splitting the set into training, validation, and test sets in a 70-15-15 ratio. To improve the generalization capacity of the model, random flipping, rotation, and scaling data augmentation techniques were applied during training to improve the database diversity[\[14\]](#). Subsequently, we set up the YOLOv5, choosing the variants with higher accuracy, YOLOv5m for better speed and high accuracy and YOLOv5l for accuracy based with slower speed[\[18\]](#). During the training, the performance of the model was monitored using different metrics such as loss, precision, recall, and mean Average Precision (mAP) to evaluate the result of the model on the validation set.

V. Results / Key Findings

A. Model Evaluation

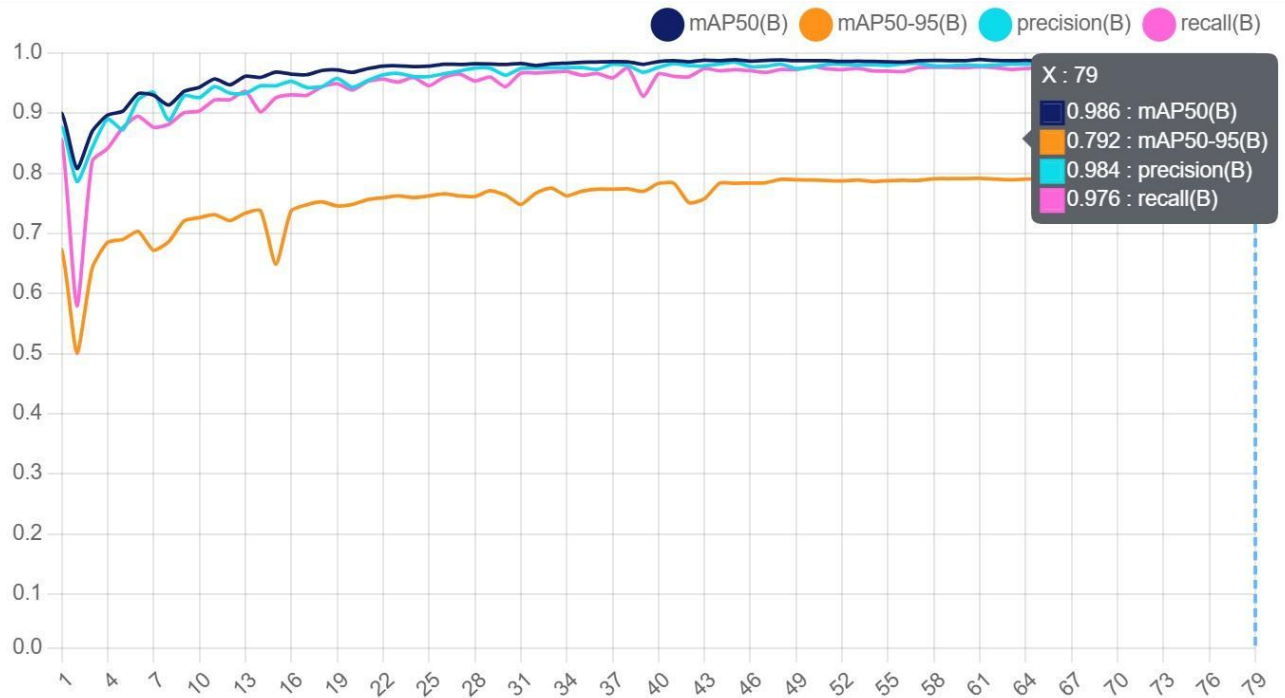
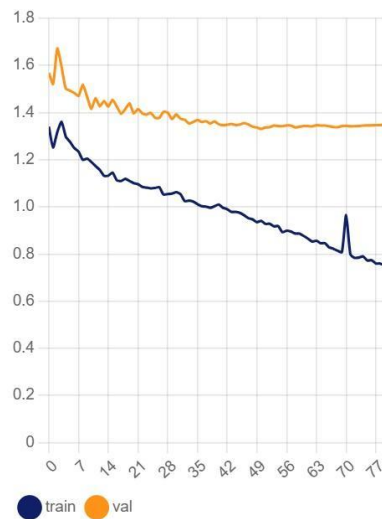


Figure 5: YOLOv5lu Metrics

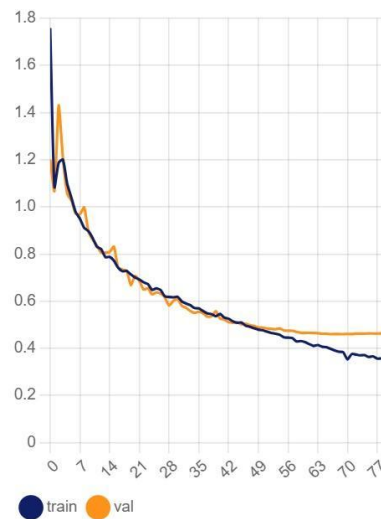
Box Loss

Difference between predicted and true boxes



Class Loss

The accuracy of the classes in each detection



Object Loss

The detected objects

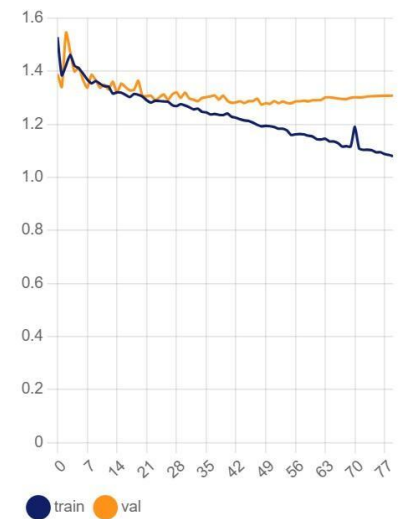


Figure 6: YOLOv5lu Box Loss, Class Loss, and Object Loss

The following will be used to gauge the model's performance:

Accuracy: Indicates how accurate the model can predict various accurate or false scenarios. This is done by comparing the proportion of accurate predictions to the total predictions. High accuracy means it is reliable, which is important in medical applications.

Precision: The ratio of the actual positive predicted values to all the predicted positive values by the model. The higher the precision, the lower the false positives, which is critical because it reduces the false diagnosis of healthy patients.

Recall: The model's ability to find all the relevant scenarios; it states the ratio of true positive cases to actual positive cases. If recall is high, the model will not omit any of the actual lung cases, which is necessary for medical applications.

These metrics will aid in assessing the lung classification model's dependability[11]. A well-performing model will exhibit high precision and accuracy, which qualifies it for real-time applications[10]. By carefully assessing these performance metrics, we can ensure that the model provides consistent insights in clinical settings, ultimately assisting medical professionals in identifying and handling lung conditions.

B. Usability Test Results

Usability Testing with Radiologist Annotations

To evaluate the practical clinical relevance of LungVision-AI, a usability test was conducted involving a licensed medical adviser specializing in radiology. The radiologist was asked to annotate a series of CT scan images by manually encircling detected lung nodules and providing detailed descriptions as a radiological report would entail. These annotations were cross-referenced against the outputs generated by the deployed YOLOv5 models through our web-based application.

The usability testing procedure included:

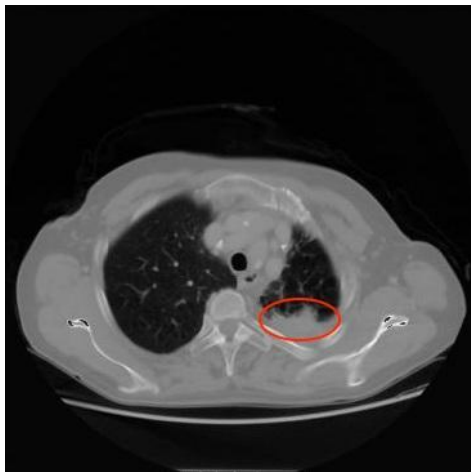
- Uploading selected CT scan images to the deployed web application.
- Comparing the model's detection bounding boxes with the radiologist's annotations.
- Analysing discrepancies, overlaps, and model prediction accuracy based on radiological descriptions.

Summary of Findings

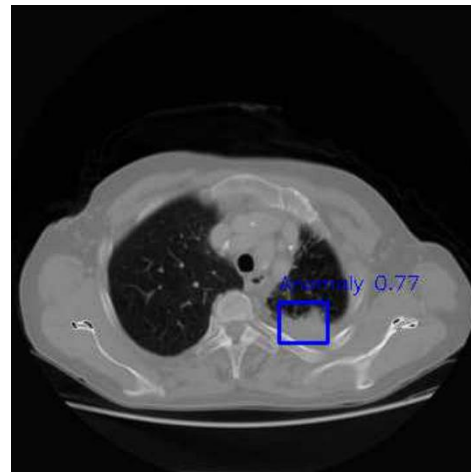
- All 15 annotated CT scan images were uploaded for testing. The model detected nodules correctly in 12 images. Three images had errors — two with no detections, and one with multiple lesions where only one was detected. This gives an 80% success rate, with misses mostly in dense or complex lung areas.

- The correctly detected nodules (true positives) had **confidence scores of 70% and above**, aligning with the **industry standard for AI object detection** in medical imaging. Misses were mainly observed in dense or complex lung areas.
- The radiologist highlighted that while the model effectively identifies lung nodules, it would be beneficial to integrate lesion descriptions to differentiate between benign and malignant features, as not all lung lesions are cancerous. Including such diagnostic insights could further support clinical decision-making. Overall, the radiologist found the interface to be straightforward and easy to use, noting that the application offers a user-friendly experience suitable for clinical workflows.

Sample Annotations (Sample_Image_14):



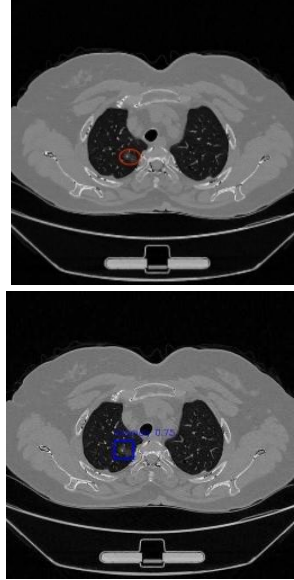
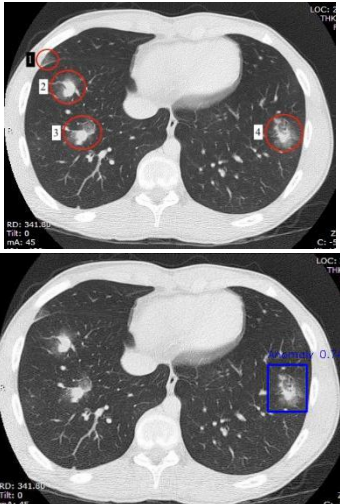
Radiologist's Annotation [\[28\]](#)

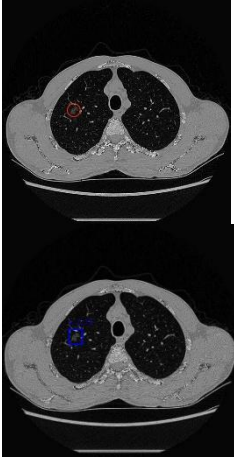
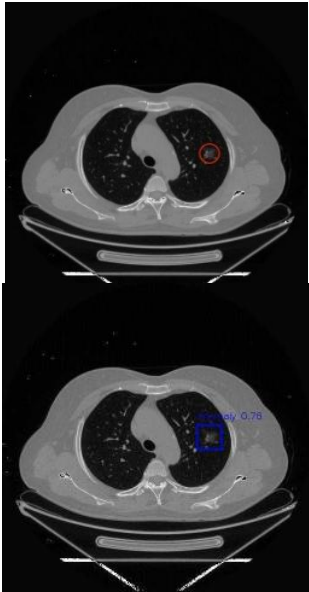


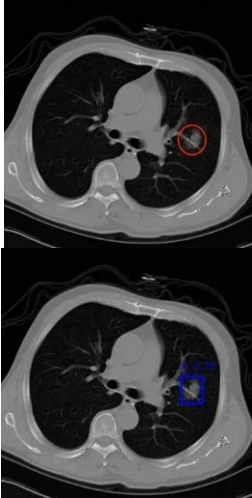
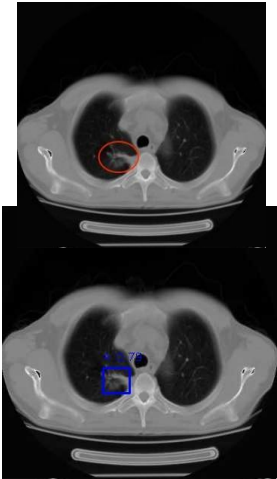
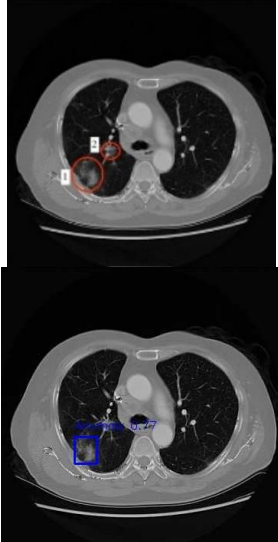
Model Prediction

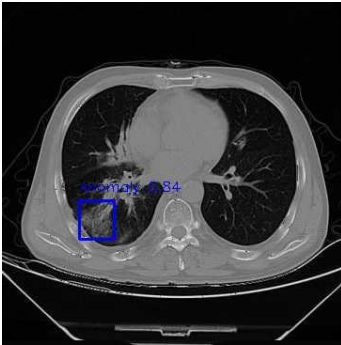
Radiologist Annotation vs Model Prediction (with Radiologist's Description):

Radiologist Annotation [28] vs Model Prediction	Radiologist's Description	Model Accuracy
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	Small non-calcified solid nodules left apicoposterior upper lobe, the smaller of the two is more medial in location [29].	Missed Detection
	Small bilobed subsolid/ground glass nodule, right posterior upper lobe [29].	Correct Detection
	Lesion 1: Perifissural ground glass opacity, bordering the right lateral middle lobe. Lesion 2: Small non-calcified solid nodule with ground glass halo, right anterior lower lobe. Lesion 3: Small non-calcified solid nodule with ground glass halo, right lateral lower lobe. Lesion 4: Small non-calcified solid nodule with ground glass halo, right lateral lower lobe.	Partial Detection

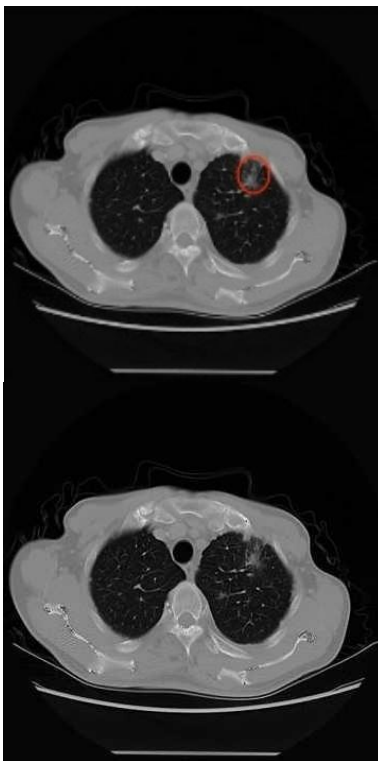
	halo, left anteromedial lower lobe [29] .	
	Small patch of ground glass opacities, right apical upper lobe [29] .	Correct Detection
	Small ill-defined subsolid/ground glass nodule, left apicoposterior upper lobe [29] .	Correct Detection

	Small lobulated subsolid/ground glass nodule left superior lingula [29].	Correct Detection
	Small ill-defined subsolid/ground glass nodule with pleural tagging, right posterior [29].	Correct Detection
	Lesion 1: Ill-defined patch of ground glass opacities, periphery of the right superior lower lobe. Lesion 2: Small subsolid/ground glass nodule, right superior lower lobe [29].	Partial Detection



Lesion 1: Patch of consolidation with air bronchogram, right lateral middle lobe.
Lesion 2: Ground glass opacities with bronchovascular thickening right lateral lower lobe [\[29\]](#).

Partial Detection



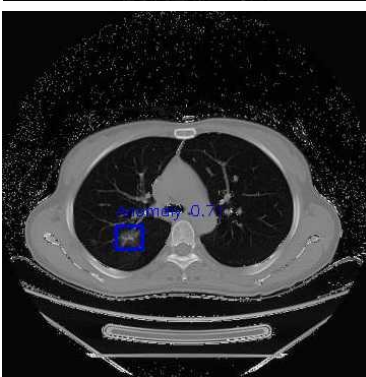
Ill-defined subsolid/ground glass nodule with pleural tagging, left anterior upper lobe [\[29\]](#).

Missed Detection



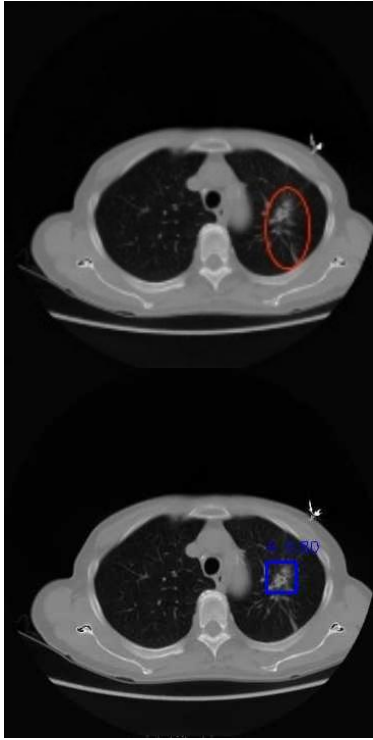
Solid lesion, medial aspect of the
left apicoposterior upper lobe
[\[29\]](#).

Correct Detection



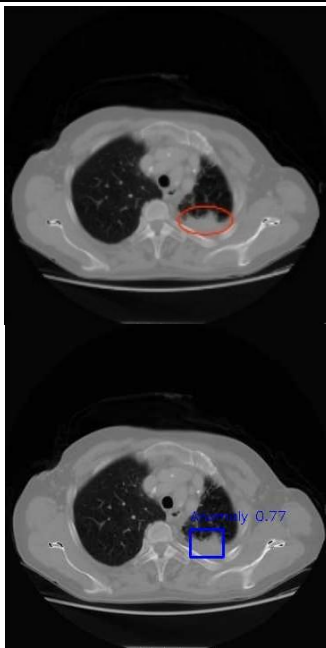
Small subsolid/ground glass
nodule, right superior lower lobe
[\[29\]](#).

Correct Detection



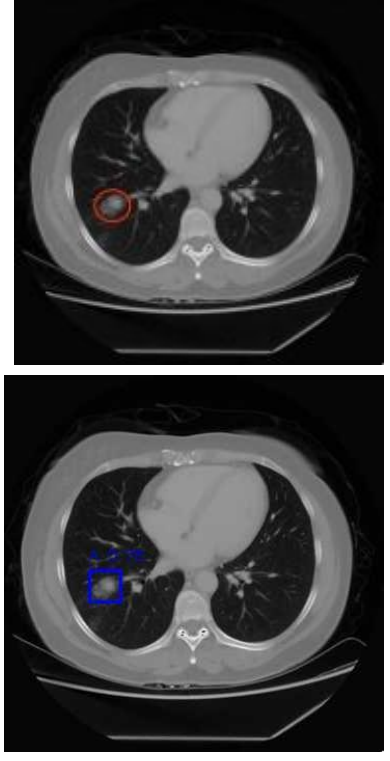
**Ill-defined subsolid lesion with
pleural tagging, left
apicoposterior upper lobe [29].**

Correct Detection



**Lobulated, mass-like left
apicopleural thickening [29].**

Correct Detection

	Small lobulated subsolid nodule, right lateral lower lobe [29].	Correct Detection
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Summary of Radiologist’s Annotation vs Model Prediction:

<i>Category</i>	<i>Count</i>	<i>Percentage</i>
<i>Total CT Scan Images</i>	15	100%
<i>Correct Detections</i>	9	60%
<i>Partial Detections</i>	3	20%
<i>Missed Detections</i>	3	20%
<i>Detection Success (Correct + Partial)</i>	12	80%

Among the 15 test cases interpreted by the radiologist, LungVision-AI accurately identified 9 lung nodules, identified 3 partially, and missed 3. This provides an overall success rate of 80%, with the majority of the failures in dense or complex anatomy areas.

C. Model Deployment

For deployment, we developed a web application with Vue.js for the frontend that is deployed on Vercel and FastAPI for the backend that is deployed on Render, combining two variants of YOLOv5 and three models. The application is designed to handle various file formats (JPG, JPEG, PNG, DICOM) with a limit of 200 MB upload size, which offers flexibility for clinical applications. When a file is submitted, the system runs images using the chosen YOLOv5 model variant and creates visualizations that display both the original input image and the output with bounding boxes around detected anomalies. Each detection has a confidence score, allowing users to evaluate the validity of each result. This deployment greatly simplifies the process for anomaly detection, providing an easy-to-use interface that needs little training while still offering high detection accuracy.

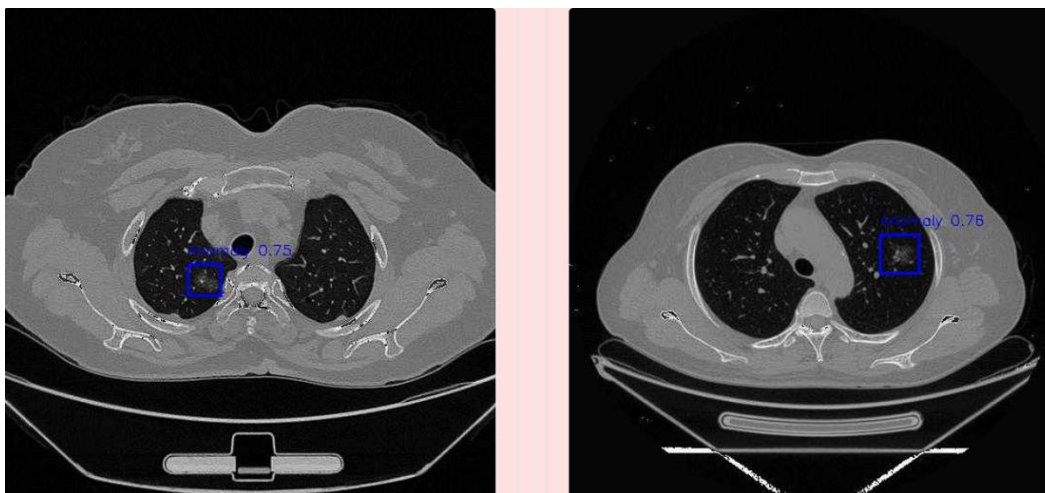


Figure 6: Deployed Website Application Result Sample

VI. Conclusion

This study developed LungVision-AI, an artificial intelligence system based on YOLOv5 architectures to assist radiologists in detecting potential lung cancer nodules from CT scan images. Through the comparison of three variant models — YOLOv5mu (medium ultra), YOLOv5lu (large ultra), and YOLOv5m6u (medium 6 ultra) — we analyzed the trade-offs between speed, computational expense, and detection accuracy to address different clinical needs. The models were trained on a robust dataset of 13,237 annotated CT images enriched through extensive preprocessing and augmentation methods to improve generalization.

Our results show that each of the three models produced exceptional detection performance. YOLOv5mu produced faster inference rates with lower computational costs, YOLOv5lu produced better accuracy at higher computational costs, and YOLOv5m6u traded larger input size gains for slightly better accuracy with a moderate

computational profile compared to YOLOv5lu. Usability testing with a boarded radiologist showed an 80% rate of detection success with failures in most cases happening in dense or anatomically complicated lung areas. Lastly, testing the models through a web-based application validated the system's usability in clinical work flows by allowing real-time or near-real-time diagnostic assistance in an intuitive user interface.

In general, LungVision-AI has the vast potential to reduce diagnostic delays, assist radiologists, and improve early lung cancer detection outcomes. Successful integration of real-time AI models into the diagnostic workflow is a major step towards clinical efficiency and improvement in patient care.

VII. Future Work

In the future, LungVision-AI will keep on focusing on making more optimization into the faster inference and less computational cost of the YOLOv5mu, YOLOv5lu and YOLOv5m6u models by applying techniques such as model pruning and quantization. The incorporation of explainable AI (XAI) ways such as Grad-CAM will allow for more explainability and further build trust among clinicians. Expansion of the system to identify multi-class—benign vs. malignant nodules—will augment its diagnostic value. Detection performance will also be enhanced in dense or anatomically complex lung regions with more advanced feature extraction techniques. Clinical validation on heterogeneous, multi-institutional data will be required to demonstrate generalizability, and inclusion of multi-modal imaging data such as PET/CT may further augment diagnostic performance. Last, real-world deployment planning will include developing secure, cloud-based platforms compliant with medical data standards to allow integration into hospital workflow.

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